

Exercise Sheet #4

Fields 1

Exercise 1: Discrete Charge Distribution

Two point charges of $+3.0 \mu\text{C}$ each are located at the points $(x, y) = (0, 2.0 \text{ m})$ and $(x, y) = (0, -2.0 \text{ m})$. Two additional point charges, each with charge q , are located at $(x, y) = (4.0 \text{ m}, 2.0 \text{ m})$ and $(x, y) = (4.0 \text{ m}, -2.0 \text{ m})$. The electric field at the point $(x, y) = (0, 0)$ due to the four charges is $\vec{E} = (4.0 \times 10^3 \hat{x}) \text{ N/C}$. Determine the value of q .

Solution

The electric field at the origin due to a point charge Q at position (x, y) is given by

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r},$$

where $\hat{r}$ is the unit vector pointing from the charge to the field point.

For the two charges at $(0, 2 \text{ m})$ and $(0, -2 \text{ m})$, the electric field components at the origin point along the y -axis. Since they are equal in magnitude and opposite in direction, they cancel each other:

$$E_y = 0 \quad (\text{net field from the two } 3.0 \mu\text{C} \text{ charges})$$

The distance to the origin of each q charge is

$$r = \sqrt{(4.0)^2 + (2.0)^2} \text{ m} = \sqrt{20} \text{ m},$$

and the electric field at the origin due to one q charge has magnitude

$$E_q = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}.$$

At the origin, the y -components of the fields from the charges at $(4 \text{ m}, 2 \text{ m})$ and $(4 \text{ m}, -2 \text{ m})$ cancel, while the x -components add.

Thus, the total field at the origin due to both q charges is

$$|E_x| = 2 \cdot E_q \cos \theta,$$

where $\cos \theta = \frac{x}{r} = \frac{4}{4.472} = 0.894$.

Given that the total field at the origin is

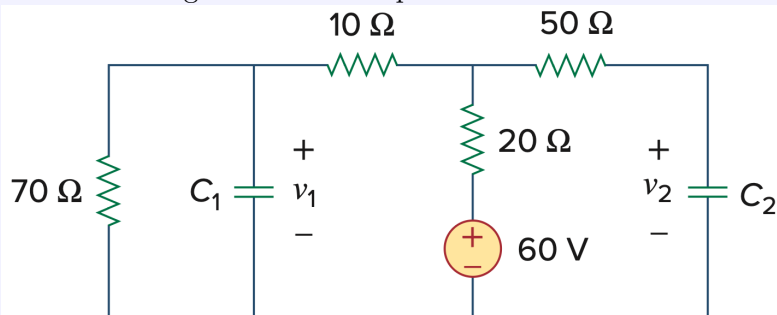
$$E_x = 4.0 \times 10^3 \text{ N/C},$$

we have

$$q = -\frac{E_x \cdot 4\pi\epsilon_0 \cdot r^2}{2 \cdot \cos \theta} = -\frac{4.0 \times 10^3 \cdot 2\pi \cdot 8.854 \times 10^{-12} \cdot 20}{0.894} \text{ C} = -5.0 \mu\text{C}.$$

Exercise 2: Capacitors in DC Steady State

Find the voltages across the capacitors in the circuit below under DC conditions.



Solution

Under DC steady-state conditions, the capacitors act as open circuits. Therefore, no current flows through C_1 or C_2 , and we can analyze only the resistive network.

Let the bottom node be the reference (0 V). The voltage at the top of the 60 V source is $V_C = 60$ V. Let V_A and V_B be the node voltages at the top of C_1 and the junction between the 10 Ω, 20 Ω, and 50 Ω resistors, respectively.

$$\begin{aligned} \text{KCL at node } A : \quad & \frac{V_A - 0}{70} + \frac{V_A - V_B}{10} + I_{C_1} = 0 \\ \Rightarrow & \frac{V_A - 0}{70} + \frac{V_A - V_B}{10} = 0 \end{aligned}$$

$$\begin{aligned} \text{KCL at node } B : \quad & \frac{V_B - V_A}{10} + \frac{V_B - 60}{20} + I_{C_2} = 0 \\ \Rightarrow & \frac{V_B - V_A}{10} + \frac{V_B - 60}{20} = 0 \end{aligned}$$

Multiply the first equation by 70:

$$V_A + 7(V_A - V_B) = 0 \quad \Rightarrow \quad 8V_A - 7V_B = 0$$

Multiply the second equation by 20:

$$2(V_B - V_A) + (V_B - 60) = 0 \quad \Rightarrow \quad -2V_A + 3V_B = 60$$

Substitute into each other:

$$\begin{aligned} -2\left(\frac{7}{8}V_B\right) + 3V_B &= 60 \\ \Rightarrow \left(-\frac{14}{8} + \frac{24}{8}\right)V_B &= 60 \\ \Rightarrow V_B &= 60 \cdot \frac{8}{10} = 48 \text{ V} \end{aligned}$$

$$V_A = \frac{7}{8}V_B = \frac{7}{8} \cdot 48 = 42 \text{ V} = v_1$$

Finally, let V_D be the trivial node (just one input and one output) voltage between the top of C_2 and the 50 Ω resistor.

$$\text{KCL at node } D : \quad \frac{V_D - V_B}{50} + I_{C_2} = 0$$

$$I_{C_2} = 0 \text{ (DC steady state)} \Rightarrow \frac{V_D - V_B}{50} = 0 \Rightarrow V_D = V_B \Rightarrow v_2 = V_B$$

$$v_1 = 42 \text{ V}, \quad v_2 = 48 \text{ V}$$

Exercise 3: Plate Capacitor

We want to design a capacitor with capacitance $C = 1 \mu\text{F}$. Compute the length required for rectangular plates of 2 cm width if the dielectric is polyester ($\epsilon_r = 3.4$) of $15 \mu\text{m}$ thickness.

Solution

For a parallel-plate capacitor

$$C = \epsilon_0 \epsilon_r \frac{A}{d},$$

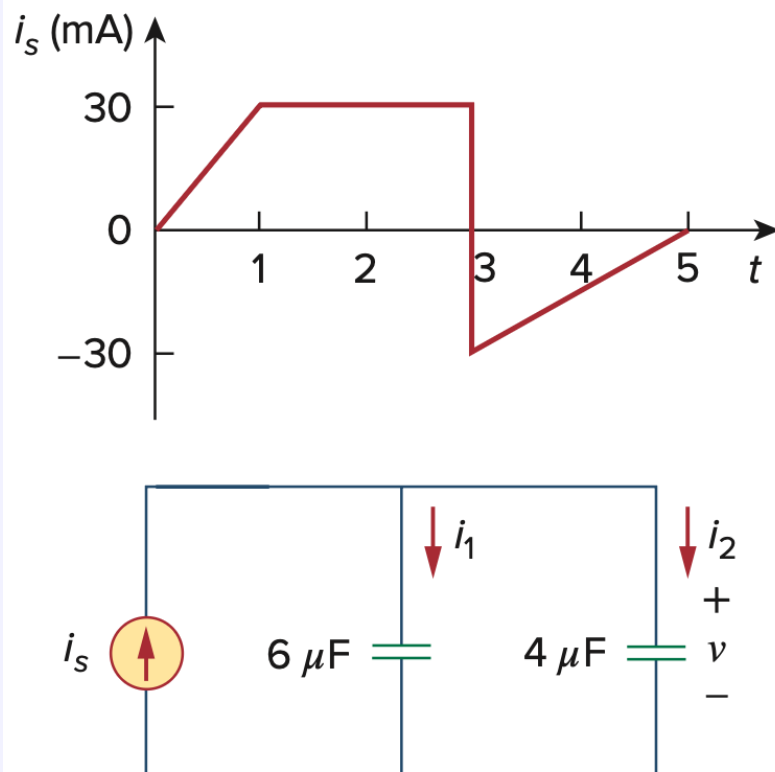
where A is the plate area and $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$ is the vacuum permittivity. For rectangular plates $A = wL$. Solving for L gives

$$L = \frac{C d}{\epsilon_0 \epsilon_r w} = \frac{1 \times 10^{-6} \cdot 15 \times 10^{-6}}{8.85 \times 10^{-12} \cdot 3.4 \cdot 0.02} \text{ m} = 24.93 \text{ m}.$$

Remarks: This large length is a consequence of the relatively small plate width and the modest relative permittivity; achieving $1 \mu\text{F}$ with a simple parallel-plate geometry and a thin polyester dielectric requires a very large plate area. Practical capacitors obtain large capacitances by using multiple layers, rolled foils, much higher-permittivity dielectrics, or much smaller effective separations (thin films, electrolytic constructions, etc.).

Exercise 4: Time-Dependent Capacitor Voltage and Current

If $v(0) = 0$, find $v(t)$, $i_1(t)$, and $i_2(t)$ in the circuit below.



Solution

$$i = C \frac{dv}{dt} \Rightarrow v = \frac{\int i \, dt}{C}$$

Here,

$$v(t) = \frac{\int i_s \, dt}{C_{eq}},$$

with the equivalent capacitance

$$C_{eq} = 6 \, \mu\text{F} + 4 \, \mu\text{F} = 10 \, \mu\text{F}.$$

For $0 < t < 1$ s:

$$v(t) = \frac{\int 30t \, dt \, \text{mA} \cdot \text{s}}{10 \, \mu\text{F}} = \frac{15t^2 + \text{const}}{10} \, \text{kV}$$

$$v(0) = 0$$

$$\Rightarrow \text{const} = 0$$

$$\Rightarrow v(t) = 1.5t^2 \, \text{kV}$$

For $1 < t < 3$ s:

$$v(t) = \frac{\int 30 \, dt}{10} \, \text{kV} = \frac{30t + \text{const}}{10} \, \text{kV}$$

$$v(1 \, \text{s}) = 1.5 \cdot 1^2 \, \text{kV} = 1.5 \, \text{kV}$$

$$\Rightarrow \text{const} = 1.5 \cdot 10 - 30 \cdot 1 = -15$$

$$\Rightarrow v(t) = (3t - 1.5) \, \text{kV}$$

For $3 < t < 5$ s:

$$v(t) = \frac{\int(\frac{30}{2}t - 75) dt}{10} \text{ kV} = \frac{7.5t^2 - 75t + \text{const}}{10} \text{ kV}$$

$$v(3\text{ s}) = [1.5t^2]_0^1 \text{ kV} + [3t - 1.5]_1^3 \text{ kV} = (1.5 - 0) \text{ kV} + (7.5 - 1.5) \text{ kV} = 7.5 \text{ kV}$$

$$\Rightarrow \text{const} = 7.5 \cdot 10 - 7.5 \cdot 9 + 75 \cdot 3 = 232.5$$

$$\Rightarrow v(t) = (0.75t^2 - 7.5t + 23.25) \text{ kV}$$

The currents through each capacitor are:

$$i_1 = C_1 \frac{dv}{dt}; \quad i_2 = C_2 \frac{dv}{dt}$$

For $0 < t < 1$ s:

$$i_1 = 6 \mu\text{F} \cdot 3t \text{ kV/s} = 18t \text{ mA}$$

$$i_2 = 4 \mu\text{F} \cdot 3t \text{ kV/s} = 12t \text{ mA}$$

For $1 < t < 3$ s:

$$i_1 = 6 \mu\text{F} \cdot 3 \text{ kV/s} = 18 \text{ mA}$$

$$i_2 = 4 \mu\text{F} \cdot 3 \text{ kV/s} = 12 \text{ mA}$$

For $3 < t < 5$ s:

$$i_1 = 6 \mu\text{F} \cdot (1.5t - 7.5) \text{ kV/s} = (9t - 45) \text{ mA}$$

$$i_2 = 4 \mu\text{F} \cdot (1.5t - 7.5) \text{ kV/s} = (6t - 30) \text{ mA}$$

Exercise 5: Energy of Capacitors

Two capacitors ($25 \mu\text{F}$ and $75 \mu\text{F}$) are connected to a 100 V source. Find the energy stored in each capacitor if they are connected in:

- a) parallel
- b) series

Solution

- a) The equivalent capacitance for parallel connection is

$$C_{eq} = C_1 + C_2 = 25 \mu\text{F} + 75 \mu\text{F} = 100 \mu\text{F}.$$

The energy stored in each capacitor is

$$E_1 = \frac{1}{2} \cdot C_1 \cdot V^2 = \frac{1}{2} \cdot 25 \times 10^{-6} \text{ F} \cdot (100 \text{ V})^2 = 125 \text{ mJ},$$

$$E_2 = \frac{1}{2} \cdot C_2 \cdot V^2 = \frac{1}{2} \cdot 75 \times 10^{-6} \text{ F} \cdot (100 \text{ V})^2 = 375 \text{ mJ}.$$

- b) The equivalent capacitance for series connection is

$$C_{eq} = \frac{C_1 \cdot C_2}{C_1 + C_2} = \frac{25 \mu\text{F} \cdot 75 \mu\text{F}}{25 \mu\text{F} + 75 \mu\text{F}} = 18.75 \mu\text{F}.$$

The voltage across each capacitor is

$$V_1 = \frac{Q}{C_1} = \frac{C_{eq} \cdot V}{C_1} = \frac{18.75 \mu\text{F} \cdot 100 \text{ V}}{25 \mu\text{F}} = 75 \text{ V},$$

$$V_2 = \frac{Q}{C_2} = \frac{C_{eq} \cdot V}{C_2} = \frac{18.75 \mu\text{F} \cdot 100 \text{ V}}{75 \mu\text{F}} = 25 \text{ V}.$$

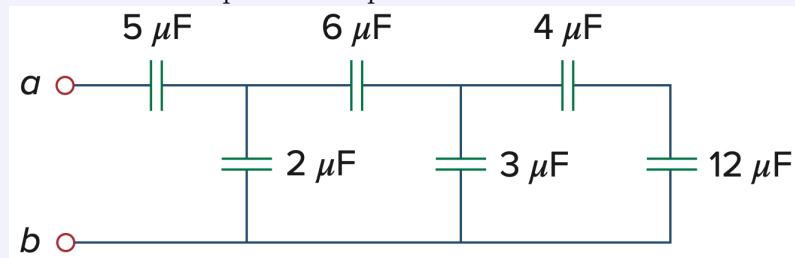
The energy stored in each capacitor is

$$E_1 = \frac{1}{2} \cdot C_1 \cdot V_1^2 = \frac{1}{2} \cdot 25 \times 10^{-6} \text{ F} \cdot (75 \text{ V})^2 = 70.31 \text{ mJ},$$

$$E_2 = \frac{1}{2} \cdot C_2 \cdot V_2^2 = \frac{1}{2} \cdot 75 \times 10^{-6} \text{ F} \cdot (25 \text{ V})^2 = 23.44 \text{ mJ}.$$

Exercise 6: Series and Parallel Capacitors

Determine the equivalent capacitance at terminals a-b of the circuit below.



Solution

$$C_{12,4,3} = \frac{12 \mu\text{F} \cdot 4 \mu\text{F}}{12 \mu\text{F} + 4 \mu\text{F}} + 3 \mu\text{F} = 6 \mu\text{F}$$

$$C_{12,4,3,6,2} = \frac{C_{4,12,3} \cdot 6 \mu\text{F}}{C_{4,12,3} + 6 \mu\text{F}} + 2 \mu\text{F} = 5 \mu\text{F}$$

$$C_{a-b} = \frac{C_{12,4,3,6,2} \cdot 5 \mu\text{F}}{C_{12,4,3,6,2} + 5 \mu\text{F}} = 2.5 \mu\text{F}$$

Exercise 7: Magnetic Force

In a straight, current-carrying section of a conductor there is a current element

$$I\vec{l}$$

with

$$I = 2.7 \text{ A}, \quad \vec{l} = (3.0 \hat{x} + 4.0 \hat{y}) \text{ cm}.$$

It is surrounded by a uniform magnetic field

$$\vec{B} = 1.3 \hat{x} \text{ T}.$$

Calculate the force acting on the section of the conductor.

Solution

Magnetic force:

$$\vec{F} = I(\vec{l} \times \vec{B})$$

Compute the cross product:

$$(\vec{l} \times \vec{B})_x = l_y B_z - l_z B_y = 0.04 \cdot 0 - 0 \cdot 0 = 0,$$

$$(\vec{l} \times \vec{B})_y = l_z B_x - l_x B_z = 0 \cdot 1.3 - 0.03 \cdot 0 = 0,$$

$$(\vec{l} \times \vec{B})_z = l_x B_y - l_y B_x = 0.03 \cdot 0 - 0.04 \cdot 1.3 = -0.052.$$

$$\Rightarrow \vec{l} \times \vec{B} = -0.052 \hat{z} \text{ m} \cdot \text{T}$$

Therefore:

$$\vec{F} = 2.7 \text{ A} \cdot (-0.052) \hat{z} \text{ m} \cdot \text{T} = -0.14 \hat{z} \text{ N}$$

Exercise 8: Lorentz Force

A velocity selector operates with a magnetic field of $B = 0.28 \text{ T}$ perpendicular to an electric field of $E = 0.46 \text{ MV/m}$. How fast must a particle move in order to pass through the selector without deflection?

Solution

For no deflection, the electric and magnetic forces must balance:

$$qE = qvB$$

$$\Rightarrow v = \frac{E}{B} = \frac{0.46 \times 10^6 \text{ V/m}}{0.28 \text{ T}} = 1.64 \times 10^6 \text{ m/s}$$

Exercise 9: Faraday's Law: Moving Conductor

A rectangular conducting loop has width $l = 0.2 \text{ m}$ (the side that crosses the field boundary) and height $h = 0.1 \text{ m}$. The loop lies initially entirely inside a region of uniform magnetic field B directed *into* the page, with magnitude $B = 0.8 \text{ T}$. The loop is pulled to the right with constant speed $v = 0.5 \frac{\text{m}}{\text{s}}$, so that it leaves the magnetic region.

- Calculate the magnitude of the induced voltage in the loop while it is partially leaving the magnetic region.
- State the direction of the induced current (clockwise or counterclockwise) as seen by the observer, and justify using Lenz's law.

Solution

Magnetic flux through the portion of the loop inside the field is $\Phi = B A_{\text{in}}$. As the loop leaves, A_{in} decreases, so $d\Phi/dt < 0$. Faraday's law gives

$$V = -\frac{d\Phi}{dt},$$

and for the straight side moving at speed v the magnitude of the induced voltage is

$$|V| = B \cdot l \cdot v.$$

a) Magnitude:

$$|V| = B \cdot l \cdot v = 0.8 \text{ T} \cdot 0.2 \text{ m} \cdot 0.5 \frac{\text{m}}{\text{s}} = 0.08 \text{ V}$$

b) Direction: The magnetic flux (into the page) is decreasing as the loop leaves. By Lenz's law the induced current will produce a magnetic field that tries to maintain the lost flux — i.e., it will produce a magnetic field *into* the page. The conducting loop can be interpreted as a mini electromagnet (coil). So, using the right-hand rule (curl the fingers in the direction of current; thumb gives the magnetic field), a magnetic field into the page corresponds to a clockwise current as seen by the observer. Thus the induced current is clockwise.

Exercise 10: Faraday's Law: Time-varying Field

A circular conducting loop of radius $r = 5 \text{ cm}$ lies in a region where the magnetic field perpendicular to the loop is increasing in time according to

$$B(t) = 0.012 t \text{ T} \quad \text{with } t \text{ in seconds.}$$

The loop resistance is $R = 2 \Omega$.

- Find the magnitude of the induced voltage at $t = 4 \text{ s}$.
- Find the induced current magnitude at that time and specify its sense (clockwise or counterclockwise) if B points *out* of the page.

Solution

Magnetic flux through the loop is $\Phi(t) = B(t) \cdot \pi r^2$. Faraday's law gives the induced voltage magnitude

$$|V| = \left| \frac{d\Phi}{dt} \right| = \pi r^2 \cdot \left| \frac{dB}{dt} \right|.$$

Here dB/dt is constant because $B(t) = 0.012 t \text{ T}$.

a) Magnitude: Radius $r = 0.05 \text{ m}$ and $dB/dt = 0.012 \text{ T/s}$. Thus

$$|V| = \pi \cdot 0.05^2 \cdot 0.012 \text{ V} \approx 9.42 \times 10^{-5} \text{ V}.$$

b) Induced current and direction: According to Ohm's law

$$I = |V|/R = \frac{9.42 \times 10^{-5}}{2} \text{ A} \approx 4.71 \times 10^{-5} \text{ A}.$$

Direction: since B is increasing *out* of the page, the induced magnetic field must oppose the increase and therefore be *into* the page. A magnetic field into the page corresponds to a **clockwise** induced current (viewed by the observer).